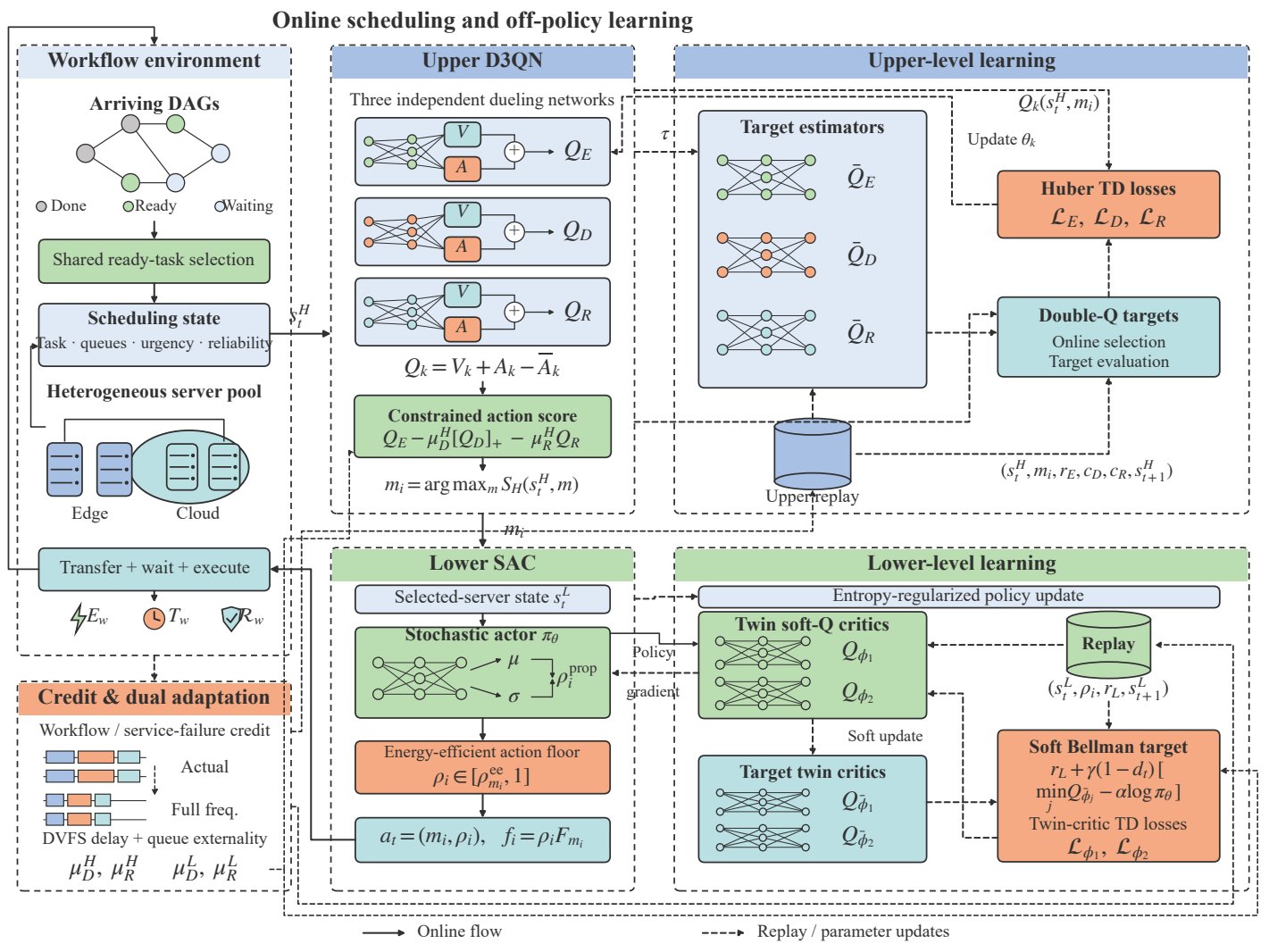

Online scheduling and off-policy learning
Workflow environment
Arriving DAGs
Done
Ready
Waiting
Shared ready-task selection
Scheduling state
Task · queues · urgency · reliability
Heterogeneous server pool
Edge
Cloud
Transfer + wait + execute
E_w
T_w
$\mathcal{R}_w$
Credit & dual adaptation
Workflow / service-failure credit
Actual
Full freq.
DVFS delay + queue externality
$\mu_D^H,\ \mu_R^H$
$\mu_D^L,\ \mu_R^L$
s_t^H
Upper D3QN
Three independent dueling networks
Q_E
Q_D
Q_R
$Q_k = V_k + A_k - \overline{A}_k$
Constrained action score
$Q_E - \mu_D^H[Q_D]_+ - \mu_R^H Q_R$
$m_i = \arg\max_m S_H(s_t^H, m)$
m_i
Lower SAC
Selected-server state s_t^L
Stochastic actor π_θ
μ
σ
ρ_i^{prop}
Energy-efficient action floor
$\rho_i \in [\rho_{m_i}^{\text{ee}}, 1]$
$a_t = (m_i, \rho_i),\quad f_i = \rho_i F_{m_i}$
Upper-level learning
Target estimators
$\bar{Q}_E$
$\bar{Q}_D$
$\bar{Q}_R$
τ
$Q_k(s_t^H, m_i)$
Update θ_k
Huber TD losses
$\mathcal{L}_E,\ \mathcal{L}_D,\ \mathcal{L}_R$
Double-Q targets
Online selection
Target evaluation
Upper replay
$(s_t^H, m_i, r_E, c_D, c_R, s_{t+1}^H)$
Lower-level learning
Entropy-regularized policy update
Twin soft-Q critics
Q_{ϕ_1}
Q_{ϕ_2}
Policy gradient
Replay
$(s_t^L, \rho_i, r_L, s_{t+1}^L)$
Soft update
Target twin critics
$Q_{\bar{\phi}_1}$
$Q_{\bar{\phi}_2}$
Soft Bellman target
$r_L + \gamma(1-d_t)[\min_j Q_{\bar{\phi}_j} - \alpha \log \pi_\theta]$
Twin-critic TD losses
$\mathcal{L}_{\phi_1},\ \mathcal{L}_{\phi_2}$
Online flow
Replay / parameter updates